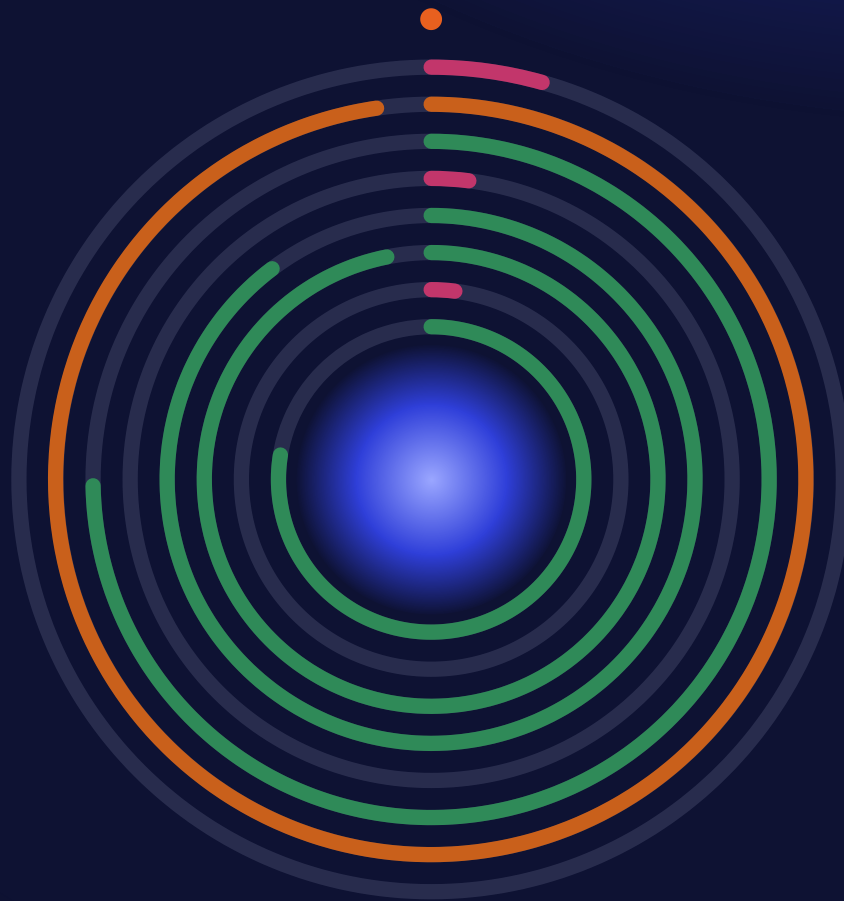


LLM Release Claims Audit

Do vendor claims survive independent measurement?



Prepared by Pawan Maheshwari, Planet AI

Edition 2026-09-25

Each ring is one claim; arc length shows how much of it held

4 of 8 vendor claims with independent data held up, but several gains depend on settings the headline leaves out or came with a quality cost.

12

claims examined across 4 releases

50%

of claims with independent data verified

4

claims fell short or held only in part

Exhibit 1

Verdicts across all claims



Source: Assay analysis of vendor release notes against independent measurements; see chapters for sources.

Key findings

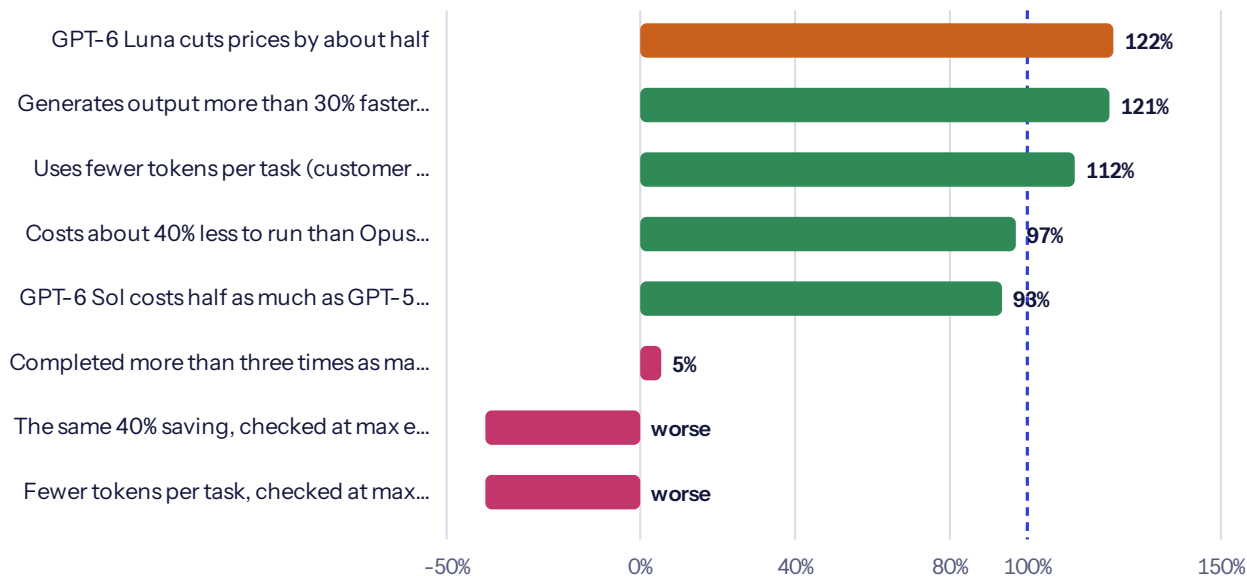
- **Anthropic delivered.** Generates output more than 30% faster than Opus 5: 121% of the claim held on independent data.^[1, 2]
- **Anthropic delivered.** Uses fewer tokens per task (customer reports range from 20% to 66% fewer): 112% of the claim held on independent data.^[1, 2]
- **Anthropic fell short outside the stated conditions.** Fewer tokens per task, checked at max effort: measured -63% against +20% claimed.^[5]
- **Cheaper is not always equal.** Intelligence Index fell from 43 to 41, so the gain came with lower measured quality.^[7]
- **4 claims have no independent data yet.** Live test packs are ready so these can be measured on your own workload.

What leaders should do

Test at the effort level you will actually deploy, budget on cost per correct answer rather than price per token, treat any single-benchmark claim as a hypothesis until it survives a noise check, and re-run the audit on every release.

Headline savings hold at the settings vendors quote, and several reverse outside them.

Exhibit 2
Share of each claim delivered on independent data



100% means the measured improvement equals the claim. Bars are coloured by verdict. Source: independent measurements cited in each chapter; Assay analysis.

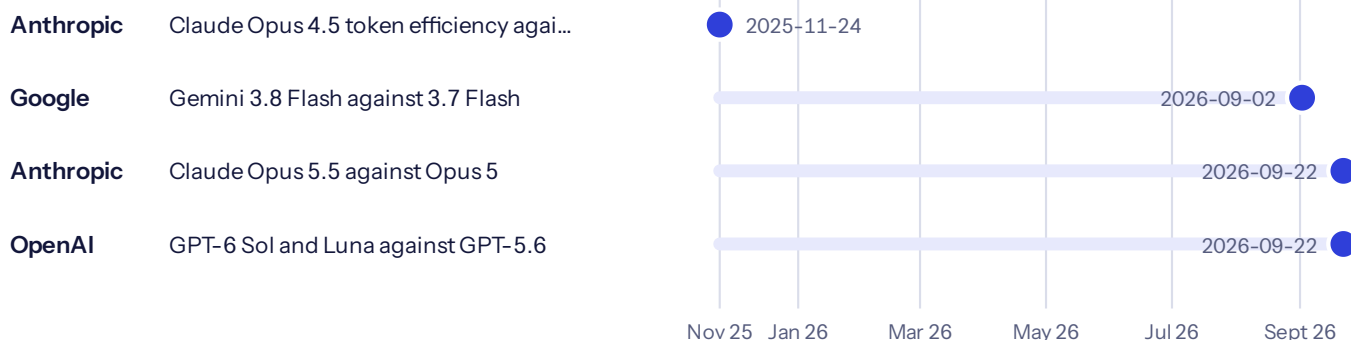
Exhibit 3
Verdict matrix by claim type

Release	Cost	Speed	Tokens	Quality
Anthropic Claude Opus 5.5 against Opus 5	Hallmarked Fool's gold	Hallmarked	Hallmarked Fool's gold	-
OpenAI GPT-6 Sol and Luna against GPT-5.6	Hallmarked Plated	-	-	Unassayed
Google Gemini 3.8 Flash against 3.7 Flash	-	-	Unassayed	Fool's gold
Anthropic Claude Opus 4.5 token efficiency against Sonnet 4.5	Unassayed	-	Unassayed	-

4 of 6 model upgrades cut the output price, by up to 58%; 2 held or raised it.

Exhibit 4

Release timeline

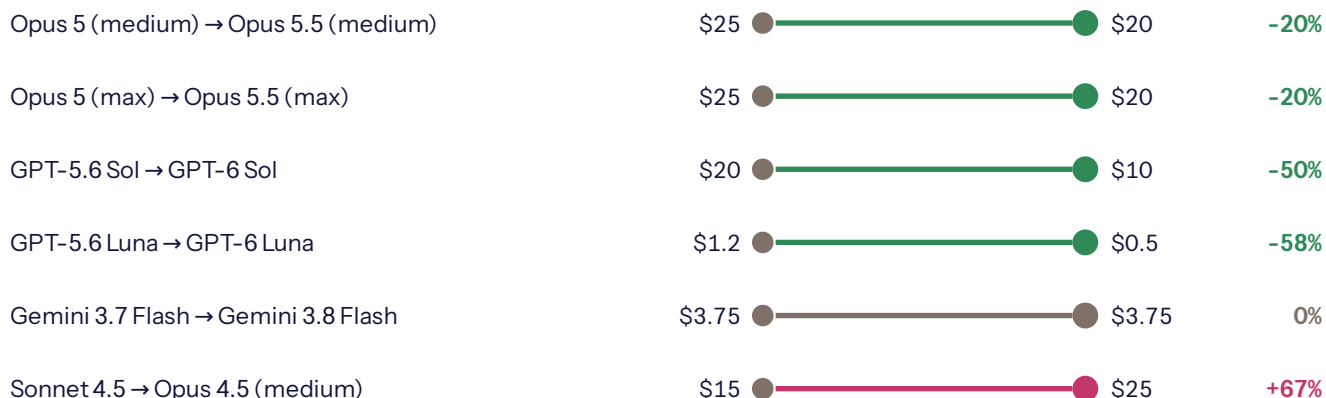


Blue: reviewed scenarios. Orange: drafts awaiting review.

Exhibit 5

Output price per release

List output price, \$ per 1M tokens, old → new



List prices from the vendor pricing pages cited in each chapter, at the time of release.

Claude Opus 5.5 against Opus 5

Released 2026-09-22. Sources checked 2026-09-25.

What the vendor said

Anthropic cut Opus prices by 20% and says Opus 5.5 costs about 40% less to run than Opus 5 at default effort, while generating output more than 30% faster. Independent measurements support the claim at the default medium effort. At max effort the story reverses: Opus 5.5 writes far more tokens and costs about the same per task.^[6]

- Costs about 40% less to run than Opus 5 on typical workloads at default effort
- The same 40% saving, checked at max effort (checked outside stated conditions)
- Generates output more than 30% faster than Opus 5
- Uses fewer tokens per task (customer reports range from 20% to 66% fewer)
- Fewer tokens per task, checked at max effort (checked outside stated conditions)

Models and list prices

Model	\$ in	\$ out
Opus 5 (medium) claude-opus-5	5	25
Opus 5.5 (medium) claude-opus-5-5	4	20
Opus 5 (max) claude-opus-5	5	25
Opus 5.5 (max) claude-opus-5-5	4	20

Per 1M tokens.

So what

The claim holds under the vendor's stated conditions but not outside them. Pin the setting you deploy in your contract or configuration, and test at that setting.

Exhibit 6

What independent data shows

Costs about 40% less to run than Opus 5 on typical workloads at default effort^[1, 2]

Anthropic. Claimed +40% on cost per task.



Delivered 97% of the claimed improvement. Intelligence Index: 45 → 51. Published: 2.19 → 1.34 \$ per index task.

The same 40% saving, checked at max effort^[3, 4]

Anthropic. Claimed +40% on cost per task.



It moved the wrong way. Intelligence Index: 51 → 58. Published: 5.86 → 5.98 \$ per index task. Anthropic's claim is for default effort. This row shows what happens if you run both at max.

Generates output more than 30% faster than Opus 5^[1, 2]

Anthropic. Claimed +30% on output speed (tokens/s).



Delivered 121% of the claimed improvement. Published: 55 → 75 output tokens per second.

Uses fewer tokens per task (customer reports range from 20% to 66% fewer)^[1, 2]

Anthropic, quoting customers. Claimed +20% on output tokens (incl. reasoning).



Delivered 112% of the claimed improvement. Published: 49 → 38 million tokens across the index. Tested against the lowest customer figure, 20%.

Fewer tokens per task, checked at max effort^[5]

Anthropic, quoting customers. Claimed +20% on output tokens (incl. reasoning).



It moved the wrong way. Published: 73 → 119 thousand output tokens per task. The token claim is not specific to max effort. This row shows the setting where it breaks.

What else the data shows

- Per-token prices fell 20% (\$5/\$25 to \$4/\$20) and cache reads fell 60% (\$0.50 to \$0.20), so part of the saving is pricing, not efficiency.^[6, 5]
- Thinking can no longer be switched off on Opus 5.5, so pipelines that ran Opus 5 without thinking lose that cheaper option.^[6]

Live test pack not yet run for this release.

GPT-6 Sol and Luna against GPT-5.6

Released 2026-09-22. Sources checked 2026-09-25.

What the vendor said

OpenAI halved token prices for its mid and low tiers. Independent measurement confirms cost per task roughly halves, driven entirely by price: both models use slightly more output tokens than before, and Luna's benchmark score dropped two points.^[9]

- GPT-6 Sol costs half as much as GPT-5.6 Sol
- GPT-6 Luna cuts prices by about half
- Sol makes about half as many factual mistakes as GPT-5.6 Sol

Models and list prices

Model	\$ in	\$ out
GPT-5.6 Sol gpt-5.6-sol	4	20
GPT-6 Sol gpt-6-sol	2	10
GPT-5.6 Luna gpt-5.6-luna	0.2	1.2
GPT-6 Luna gpt-6-luna	0.1	0.5

Per 1M tokens.

So what

The saving is real, but part of it came with lower measured quality. Check your own quality bar before switching.

Exhibit 7

What independent data shows

GPT-6 Sol costs half as much as GPT-5.6 Sol^[7]

OpenAI. Claimed +50% on cost per task.



Delivered 93% of the claimed improvement. Intelligence Index: 47 → 48. Published: 1.99 → 1.06 \$ per index task. OpenAI states the cut per token; this checks it per completed task.

GPT-6 Luna cuts prices by about half^[7]

OpenAI. Claimed +50% on cost per task.



Delivered 122% of the claimed improvement. Published: 0.18 → 0.07 \$ per index task. Intelligence Index fell from 43 to 41, so the gain came with lower measured quality.



Sol makes about half as many factual mistakes as GPT-5.6 Sol

OpenAI. The claim has no figure Assay can convert into a testable number.

OpenAI's figure comes from its own set of user-flagged conversations. Artificial Analysis reports its knowledge and hallucination index rising from 22 to 27, which points the same way but measures something different. Test it with a factual task set of your own.

What else the data shows

- The saving comes from price: output tokens per task rose slightly, from about 29k to 31k for Sol and 41k to 51k for Luna.^[8]
- Independent evals show regressions too: Sol dropped about 100 Elo on GDPval-AA v2.1, a benchmark of economically valuable tasks.^[8]
- Prompts above 272K input tokens are billed at higher long-context rates, so very long-context workloads save less.^[9]

Live test pack not yet run for this release.

Gemini 3.8 Flash against 3.7 Flash

Released 2026-09-02. Sources checked 2026-09-25.

What the vendor said

Same price, same limits, three weeks apart. Google leads with a customer saying 3.8 Flash completed more than three times as many tasks. Independent indexes show a real but much smaller gain in agentic work, flat coding, and a slower first token.^[12]

- Completed more than three times as many tasks as 3.7 Flash
- Agentic video understanding uses up to 88% fewer tokens on long-form content

Models and list prices

Model	\$ in	\$ out
Gemini 3.7 Flash gemini-3.7-flash	0.75	3.75
Gemini 3.8 Flash gemini-3.8-flash	0.75	3.75

Per 1M tokens.

So what

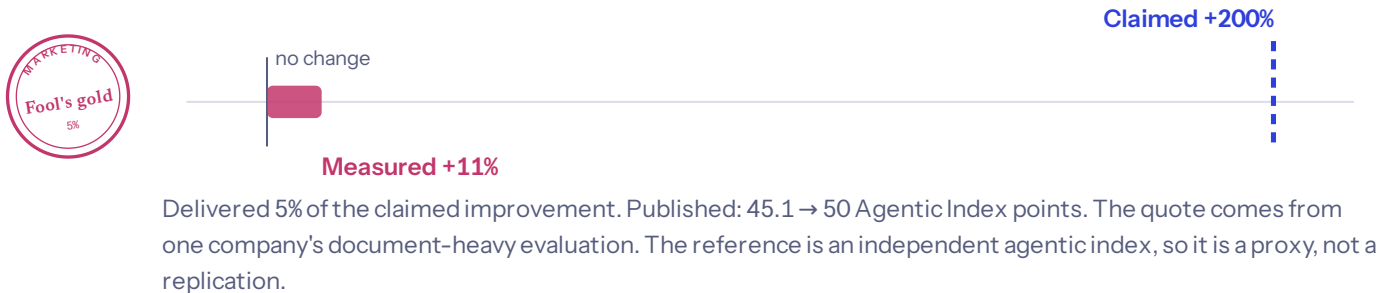
The headline figure comes from a single customer's evaluation; independent data shows a much smaller gain. Treat it as an upper bound and measure on your own workload.

Exhibit 8

What independent data shows

Completed more than three times as many tasks as 3.7 Flash^[12]

Customer quote on Google's Flash page. Claimed +200% on benchmark score (relative).



Agentic video understanding uses up to 88% fewer tokens on long-form content

Google. Claimed +88% on total tokens.

No independent measurement found. Build a long-video task set and run it live; note the claim says up to 88%, so a typical figure will be lower.

What else the data shows

- Time to first token got worse, from 12.0 to 13.2 seconds, even though streaming speed rose from 279 to 299 tokens per second.^[12]
- The independent coding index barely moved (76.1 to 76.3), while the overall Intelligence Index rose from 56 to 59.^[12]
- Both models are on introductory pricing of \$0.75/\$3.75 that rises to \$1.50/\$7.50 on 1 January 2027.^[13]

Live test pack not yet run for this release.

Claude Opus 4.5 token efficiency against Sonnet 4.5

Released 2025-11-24. Sources checked 2026-09-25.

What the vendor said

A widely repeated efficiency claim from late 2025 that was measured on one benchmark under the vendor's own harness. There is no matching independent number, so this scenario is a live test pack only.^[14]

- At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens
- Cheaper per task than Sonnet 4.5 despite higher token prices

Models and list prices

Model	\$ in	\$ out
Sonnet 4.5 claude-sonnet-4-5	3	15
Opus 4.5 (medium) claude-opus-4-5-20251101	5	25

Per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 9

What independent data shows



At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens

Anthropic. Claimed +76% on output tokens (incl. reasoning).
The figure is from one coding benchmark on Anthropic's harness. The live pack tests whether the saving carries over to short, general tasks, where it is likely to be smaller.



Cheaper per task than Sonnet 4.5 despite higher token prices

Implied by the token and price claims. Claimed +20% on cost per task.
Opus 4.5 costs \$5/\$25 against Sonnet 4.5's \$3/\$15, so it only wins per task if it really writes far fewer tokens. The 20% target here is illustrative.

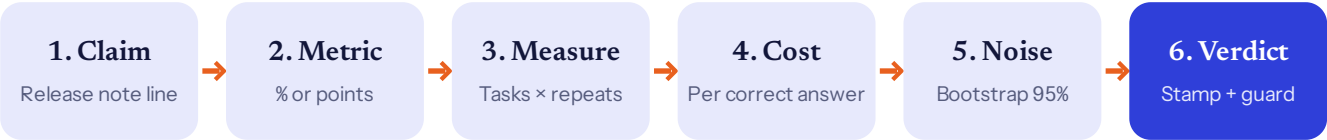
What else the data shows

- At high effort, Anthropic says Opus 4.5 beat Sonnet 4.5 by 4.3 points while using 48% fewer tokens, so the saving shrinks as effort rises.^[14]

Live test pack not yet run for this release.

Every verdict follows the same six steps, and every number traces back to a source.

Exhibit 10
The Assay validation pipeline



Rule, applied in order	Verdict
Live runs only: the 95% bootstrap range crosses zero	Unassayed
The measure moved the wrong way	Fool's gold
80% or more of the claimed improvement delivered	Hallmarked
40% to 79% delivered, or a Hallmarked gain where accuracy or the benchmark index fell by 2 points or more	Plated
Under 40% delivered	Fool's gold
No independent measurement exists	Unassayed

Improvements are relative for cost, tokens, latency and speed, and absolute points for accuracy and benchmark indexes. Cost per task counts input tokens and all output tokens, including hidden reasoning, at list price. Live runs grade every answer automatically, repeat each task, and use a paired bootstrap over tasks with 800 resamples. Reference verdicts compare two published figures and carry no noise range.

Limitations

Independent benchmarks use their own task mixes, harnesses and effort settings, which may not match yours. Vendor-reported numbers are labelled as such. Model IDs marked for confirmation, and draft scenarios extracted automatically from release notes, should be reviewed before decisions are made. This report was generated by Assay on 2026-09-25.

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